WORKED SOLUTION

STEP II 2001, Question 14

Mechanics • Coin toss

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s2-q14

The key idea

T is a weighted combination of two independent Bernoulli variables, so its mean and variance come straight from linearity of expectation and the rule $\mathrm{Var}(aX + cY) = a^2\,\mathrm{Var}(X) + c^2\,\mathrm{Var}(Y)$ for independent X, Y. The weights are chosen so that the mean is p whatever λ is: T is an *unbiased estimator* of p, and the question is asking for the best one. For the final part, the central limit theorem makes $\overline{T}$ approximately normal, and a normal variable with a fixed mean is most concentrated when its variance is smallest.

Solution

Mean and variance of T

Here X and Y are Bernoulli variables with

$$\mathrm{E}(X) = p, \quad \mathrm{Var}(X) = p(1-p), \qquad \mathrm{E}(Y) = 2p, \quad \mathrm{Var}(Y) = 2p(1-2p).$$

(For a variable taking the value 1 with probability q and 0 otherwise, $\mathrm{E} = q$ and $\mathrm{E}(\,\cdot^2) = q$, so the variance is $q - q^2 = q(1-q)$.) By linearity of expectation,

$$\mathrm{E}(T) = \lambda p + \tfrac{1}{2}(1-\lambda)\cdot 2p = \lambda p + (1-\lambda)p = p,$$

as required. Since the coins are independent, X and Y are independent, so there is no covariance term and

$$\begin{aligned}\mathrm{Var}(T) &= \lambda^2\,\mathrm{Var}(X) + \tfrac{1}{4}(1-\lambda)^2\,\mathrm{Var}(Y)\\ &= \lambda^2 p(1-p) + \tfrac{1}{2}(1-\lambda)^2 p(1-2p).\end{aligned}$$

Minimising the variance

Write $V(\lambda) = \mathrm{Var}(T)$. This is a quadratic in λ with positive leading coefficient $p(1-p) + \frac{1}{2}p(1-2p)$ (both terms are positive since $0 < p < \frac{1}{2}$), so it has a single minimum, where $V'(\lambda) = 0$:

$$V'(\lambda) = 2\lambda p(1-p) - (1-\lambda)p(1-2p) = 0.$$

Dividing by $p > 0$ and collecting the λ terms,

$$\lambda[2(1-p)+(1-2p)] = 1-2p \quad \Longrightarrow \quad \lambda = \frac{1-2p}{3-4p},$$

which is the stated value. (Here $3-4p > 1 > 0$, so there is no division by zero.)

We will also need the minimum value itself. With $a = p(1-p)$ and $c = \frac{1}{2}p(1-2p)$ we have $V = a\lambda^2 + c(1-\lambda)^2$, minimised at $\lambda = c/(a+c)$ (which is the value above), giving

$$V_{\min} = a\Big(\frac{c}{a+c}\Big)^2 + c\Big(\frac{a}{a+c}\Big)^2 = \frac{ac(a+c)}{(a+c)^2} = \frac{ac}{a+c}.$$

Now $a+c = \frac{1}{2}p[2(1-p)+(1-2p)] = \frac{1}{2}p(3-4p)$ and $ac = \frac{1}{2}p^2(1-p)(1-2p)$, so

$$V_{\min} = \frac{p(1-p)(1-2p)}{3-4p}.$$

The mean of n tosses

Let $T_1, \ldots, T_n$ be the values of T on the n tosses; they are independent, each with mean p and variance $V(\lambda)$. Then

$$\mathrm{E}(\overline{T}) = p, \qquad \mathrm{Var}(\overline{T}) = \frac{V(\lambda)}{n}.$$

Ask yourself T takes only four values. Why may we treat $\overline{T}$ as normal?

Because $\overline{T}$ is the mean of n independent, identically distributed variables with finite variance, and $n > 30$ is large, the central limit theorem says $\overline{T}$ is approximately $\mathrm{N}(p, V(\lambda)/n)$, whatever the shape of the distribution of a single T. Writing $\sigma_\lambda = \sqrt{V(\lambda)/n}$ and $Z = (\overline{T}-p)/\sigma_\lambda \approx \mathrm{N}(0,1)$,

$$\mathrm{P}(|\overline{T}-p| < b) = \mathrm{P}\Big(|Z| < \frac{b}{\sigma_\lambda}\Big) \approx \Phi\Big(\frac{b}{\sigma_\lambda}\Big) - \Phi\Big(-\frac{b}{\sigma_\lambda}\Big) = 2\Phi\Big(\frac{b}{\sigma_\lambda}\Big) - 1,$$

using the symmetry $\Phi(-z) = 1-\Phi(z)$.

Since $b > 0$ is fixed and Φ is strictly increasing, this is largest when b/σ_λ is largest, that is, when σ_λ is smallest, which is when $V(\lambda)$ is minimised. That happens at $\lambda = (1-2p)/(3-4p)$, where

$$\sigma_\lambda^2 = \frac{V_{\min}}{n} = \frac{p(1-p)(1-2p)}{(3-4p)n} = s^2.$$

So the maximum value of $\mathrm{P}(|\overline{T}-p| < b)$ as λ varies is approximately $2\Phi(b/s) - 1$, as required.

Checks

- At $\lambda = 1$, $T = X$ and the formula gives $\mathrm{Var}(T) = p(1-p)$; at $\lambda = 0$, $T = \frac{1}{2}Y$ and it gives $\frac{1}{4}\cdot 2p(1-2p)$. Both are right.

- $V_{\min}$ is smaller than both of these: it is $p(1-p)$ times $\frac{1-2p}{3-4p} < 1$. Combining the two coins beats using either alone.
- Numerically, with $p = 0.3$, $n = 100$, $b = 0.03$: the best λ is $\frac{2}{9}$, $s \approx 0.0216$, and $2\Phi(b/s) - 1 \approx 0.835$. A simulation of 40 000 runs of 100 tosses gave 0.836, while $\lambda = 1$ (coin A alone) gave only about 0.487.

Common slips

- Writing $\operatorname{Var}(\frac{1}{2}(1-\lambda)Y) = \frac{1}{2}(1-\lambda)\operatorname{Var}(Y)$: constants come out of a variance squared.
- Forgetting to say *why* minimising the variance maximises the probability: it needs Φ increasing and the mean of $\overline{T}$ being p for every λ.

Beyond STEP

This is a first taste of estimation theory: among all unbiased estimators $\lambda X + \frac{1}{2}(1-\lambda)Y$ of p, we found the one with minimum variance. The answer weights each observation in inverse proportion to its variance, the same rule used in statistics to combine measurements of differing precision (inverse-variance weighting), and the minimum $ac/(a+c)$ is the "parallel resistors" formula $1/V_{\min} = 1/a + 1/c$. Grimmett and Welsh, *Probability: An Introduction*, covers the central limit theorem behind the final step.

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